

JOINT EXAMINATION COMMITTEE VYAS MUNICIPALITY

First Learning Achievement Examination – 2083

Class: 10

Full Marks: 50

Sub: Computer Science Time: 2 Hours

Group “A” [10 × 1 = 10]

Choose the correct option from the following and write in your answer sheet.

Q.1) Which connector is primarily used for Ethernet networking cables like CAT6?

- a) USB b) RJ-45 c) HDMI d) VGA

Q.2) A topology with a single cable that connects all the workstations, servers, printers and other devices on the network.

- a) Ring topology b) Bus topology c) Star topology d) Mesh topology

Q.3) In which data communication mode does data flow in both directions, but only one direction at a time?

- a) Simplex b) Half Duplex c) Full Duplex d) Multi Duplex

Q.4) What is the main function of a DBMS?

- a) To design websites
b) To manage and store data
c) To create hardware
d) To run games

Q.5) Which data type is used to store text with variable length?

- a) INT b) CHAR c) VARCHAR d) FLOAT

Q.6) Which SQL command belongs to the Data Definition Language category?

- a) SELECT b) UPDATE c) CREATE d) INSERT

Q.7) Which of the following keys is explicitly designed to ensure every record in a table is uniquely identified?

- a) Primary Key b) Foreign Key c) Alternate Key d) Candidate Key

Q.8) Which software is commonly used for editing graphics?

- a) Windows Media Player b) Adobe Photoshop
c) VLC player d) Notepad

Q.9) A video frame is:

- a) A sound clip b) A compression method
c) A single image in a video sequence d) A graphic tool

Q.10) Flac is an example of:

- a) Video codec b) Raster format
c) Lossless audio format d) Animation tool

Group “B” [10 × 2 = 20]

Answer the following questions briefly.

Q.11) Explain the main difference between a client-server and a peer-to-peer network.

Q.12) Why is a switch considered an intelligent device compared to a hub?

Q.13) What is IP address, and why is it important?

Q.14) Define a database management system. Provide two examples of popular DBMS software.

Q.15) What is meant by a “data type” in DBMS? Name any two standard SQL data types.

Q.16) Write the general syntax of the SQL command used to create a new table.

Q.17) What does the word multimedia mean? Name the five main components of multimedia.

Q.18) Define graphics in multimedia. What does animation create.

Q.19) What is the basic unit of raster graphics? Name one software used for editing graphics.

Q.20) What is resolution in video? Define a pixel.

Group “C” [5 × 4 = 20]

Answer the following questions in Long.

Q.21) Suppose a school wants to connect computers in a computer lab. Which topology would you recommend? Justify your choice based on cost, cable length, and ease of maintenance.

Q.22) Your home internet uses a fiber-optic connection. Explain how this improves your internet experience compared to a traditional cable connection.

Q.23) Write SQL code to create a table and insert the data “customer” with the following properties, then write the SQL code to display all the customers who have a credit limit of more than 20000.

Customer ID	Name	Credit Limit	Date	Location
101	Rita	5000	2083-4-25	Damauli
102	Krishna	70000	2083-5-1	Pokhara

Q.24) Explain the concept of multimedia technology with its advantages and limitations.

Q.25) Differentiate between:

- i. Raster and vector graphics
ii. Data and informations

Best of Luck